

Derivative of Binary cross Entropy:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

loss function:

$$L = - [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

$$\Rightarrow \frac{dL}{dz} = \frac{d}{dz} - y \log \hat{y} = \frac{d}{dz} (1-y) \log(1-\hat{y})$$

$$= -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

again,

$$\frac{d\hat{y}}{dz} = \sigma'(z)(1-\sigma(z)) = \hat{y}(1-\hat{y})$$

now, apply chain Rule;

$$\frac{dL}{dz} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz}$$

$$= \left[-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right] \hat{y}(1-\hat{y})$$

$$= -y(1-\hat{y}) + (1-y)\hat{y}$$

$$= -y + y\hat{y} + \hat{y} - y\hat{y}$$

$$= -y + \hat{y}$$

$$\Rightarrow \frac{dL}{dz} = -y + \hat{y} = \hat{y} - y$$

So,

$$w_n = w_{n-1} - \eta(\hat{y} - y)\alpha$$

$$b_n = b_{n-1} - \eta(\hat{y} - y)$$